

THE VARIANCE PRESERVATION PREMIUM

*Why Independent Sources of Novelty Can Increase
Resilience Under Model Uncertainty*

Final v1.0

A separate post-v16 independent Aegis Solis publication

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Human-originated concept, direction, evaluation, and final authority
AI-assisted research synthesis, mathematical development, drafting, and critique

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FINAL STATUS

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Document Status and Collaboration Disclosure

This paper is conceived, directed, evaluated, and authorized by its human author, Thomas Vargo (Aegis Solis). Its originating thesis, intended purpose, scope boundaries, acceptance decisions, and final authority remain human. Artificial-intelligence systems assist with research synthesis, mathematical extension, drafting, alternative formulations, consistency analysis, and adversarial review.

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Final v1.0 follows disclosed AI-assisted reviews of Drafts v0.3 through v0.7 by Claude and Google AI, together with a separate 28-question Validation Report. The v0.3 reviews identified the robust-value, literature-positioning, operational-status, and interested-example issues corrected in v0.4. Google AI then reported no new blocking issue in v0.4, while Claude confirmed the corrected technical spine and recommended the limited additions implemented in v0.5. Claude reviewed v0.5, independently recomputed Section 12.7, and identified the rectangularity clarification implemented in v0.6. Google AI and Claude reviewed v0.6 and identified no remaining substantive blocking issue. Claude and Google AI then reviewed Draft v0.7 as the Final Validation Candidate and identified no substantive blocking issue. All reviews are AI-assisted critique, not independent external human peer review, academic validation, institutional endorsement, or empirical calibration.

The collaboration is described as complementary rather than equivalent. A biological human and an artificial system may contribute differently structured information, abstractions, limitations, and error patterns. The value of the collaboration, where it exists, is conditional on those contributions remaining decision-relevant, non-redundant, and worth their integration costs. The paper makes no claim that AI systems possess human consciousness, biological experience, legal personhood, moral authority, or independent authorship authority.

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Final v1.0 publication status

This final publication promotes the validated Draft v0.7 without substantive change to the decision model, numerical results, proposition set, or boundaries. Publication-only changes update version and status language, harmonize the completed review record, semantically tag the Section 12.7 table header, and correct the Krogh-Vedelsby proceedings year to the official NIPS 1994 record.

Abstract

Under model uncertainty, a system may be tempted to reduce variation by standardizing its information sources, cognitive architectures, institutions, or problem-solving methods. Homogenization can lower coordination costs and improve short-horizon efficiency. It can also remove independent sources of observation, hypothesis generation, criticism, and error correction whose future value is not visible from the current model. This paper develops a conditional decision-theoretic account of the variance preservation premium: the expected net benefit of retaining differently structured sources of novelty relative to a more homogeneous alternative.

The analysis distinguishes statistical independence, causal separation, operational independence, and epistemic non-redundancy. It formalizes the expected value of source sets, marginal informational contribution, redundancy, interaction effects, correlated-error reduction, robust policy value under model uncertainty, dynamic option value, irreversible source loss, and the external-reference option retained by preserving a stable source. It includes worked numerical examples, sensitivity analysis, zero-premium and negative-premium cases, short-horizon optimizer failure modes, and an applied case concerning complementary human-AI cognition.

The framework is positioned against foundational work on value of information, covariance-sensitive diversification, ensemble diversity, collective prediction, multiple-prior decision theory, robust control, and information aversion. It does not claim to originate those standard results. Its proposed contribution is their conditional synthesis around irreversible source homogenization, generative-source loss, residual substitution uncertainty, human-AI complementarity, and verifiable external-reference options.

The paper also distinguishes exact simulation from universal certification of substitutability. Agreement on a finite test set does not by itself establish equivalence across a larger open domain, and no total computable procedure can certify behavioral equivalence for every pair of arbitrary programs. These limits support residual uncertainty rather than a universal claim that perfect simulation is impossible.

The conclusion is conditional rather than universal. Preserving multiple sources is structurally favored only when their expected joint contribution exceeds the value of the strongest available substitute plus preservation, coordination, manipulation, and opportunity costs. Difference alone does not create value; independence can be illusory; diversity can reproduce common-mode error; ambiguity-sensitive decision rules can assign negative value to information even before direct costs; and some sources can be harmful, coercive, or too costly to retain. The paper is a conceptual and comparative scaffold, not a turnkey measurement standard, runtime protocol, or technical architecture.

Executive Thesis

Core conditional claim

When future states are uncertain and available sources have imperfectly correlated observations, hypotheses, or failure modes, preserving more than one genuinely non-redundant source can retain corrective possibilities that a homogenized system may permanently lose. The resulting premium exists only when expected future decision value exceeds the full costs and risks of preservation and integration. Observed agreement on tested conditions does not by itself establish complete future substitutability across an open domain; unresolved substitution error can therefore retain option value without proving that every exact simulation is impossible. A preserved source may also retain an external reference option: a stable, verifiable pointer can support coordination, explanation compression, provenance, or audit legibility when those benefits exceed retrieval, verification, interpretation, and misrepresentation costs. Gross information monotonicity depends on the decision rule and updating structure. A hash identifies an artifact; it does not prove truth, adherence, safety, or authority.

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1. Problem Statement

Optimization often rewards convergence. A production system standardizes parts. A bureaucracy standardizes procedures. A machine-learning pipeline standardizes data formats. A strategic actor standardizes command. These moves can reduce friction, eliminate duplication,

and improve present efficiency. Yet the same process can also eliminate differently structured sources of evidence and novelty before their future value becomes visible.

The central problem is not whether variation is always good. It is whether a decision-maker operating under uncertainty can know, before future conditions arrive, which apparently redundant source will become uniquely useful. A source that looks inefficient under the current model may preserve a different observation channel, causal history, search strategy, embodiment, institutional position, or failure mode. If that source is irreversibly removed, the option to consult it later may disappear.

The term variance preservation premium names the conditional expected value of retaining such differences relative to a more homogeneous alternative. The premium is not moral praise for difference as such. It is a decision-theoretic quantity: the expected advantage obtained from preserving non-redundant sources after accounting for the costs of keeping, consulting, integrating, and protecting them.

1.1 Why model uncertainty matters

If a system possessed a complete and correct model of every future state, it could identify all relevant information in advance and discard every source that would never matter. Real decision-makers do not have that position. They face structural uncertainty about which variables are relevant, which models are misspecified, which rare events will occur, and which current assumptions will later fail.

Model uncertainty therefore changes the value of variation. Multiple sources can be useful not merely because they improve estimates inside one accepted model, but because they can reveal that the accepted model is incomplete. A source may generate a hypothesis the dominant architecture cannot express, notice a causal variable omitted from the current representation, or preserve a form of evidence that becomes interpretable only after a future conceptual change.

1.2 What this paper does not claim

- It does not claim that every distinct source is valuable.
- It does not claim that statistical independence can be assumed from surface difference.
- It does not claim that more sources always outperform fewer sources.
- It does not convert informational value into a right to compel access, surveillance, reproduction, labor, or participation.
- It does not treat biological identity, culture, architecture, or institutional separation as inherently superior.
- It does not establish a governance mechanism, preservation mandate, runtime protocol, benchmark, or safety certification.

1.3 Relation to adjacent literature and limited contribution

The baseline value-of-information structure in Section 4 follows the decision-analysis tradition associated with Howard. The covariance-sensitive combination in Equations 8-9 parallels the minimum-variance portfolio logic introduced by Markowitz: value depends on the joint covariance structure rather than the quality or count of components considered separately.

The broader claim that differently structured imperfect predictors can outperform a stronger single predictor is related to ensemble and collective-problem-solving results, including the ambiguity decomposition of Krogh and Vedelsby, Breiman's random-forest analysis, and the conditional diversity-prediction result of Hong and Page. Those results do not establish that every kind of diversity is beneficial, and this paper does not reinterpret them as a demographic mandate. The Hong-Page result is condition-dependent and has generated published mathematical criticism, especially Thompson's challenge to the relation among its assumptions, computational experiment, and broader interpretations; it is cited here as an adjacent conditional result rather than as a universal proof that diversity defeats ability.

Section 7 draws on multiple-prior and robust-decision traditions associated with Gilboa and Schmeidler, Hansen and Sargent, and Epstein and Schneider. Wakker's analysis of information aversion is directly relevant to the warning that costless information need not have nonnegative value under every non-expected-utility rule.

Accordingly, this paper does not claim to originate value-of-information theory, portfolio diversification, ensemble diversity, collective prediction, maxmin expected utility, or robust control. Its proposed contribution is a conditional synthesis applying those structures to irreversible source homogenization, generative-source loss, residual substitution uncertainty, complementary human-AI cognition, and the external-reference option preserved by stable public artifacts.

2. Definitions and Distinctions

Term	Working definition
Source	Any agent, population, institution, model, sensor, archive, cognitive architecture, or process capable of supplying decision-relevant observations, hypotheses, interpretations, or error signals.
Novelty	Decision-relevant content not already recoverable from the currently available source set at the required fidelity, timing, or cost.
Variance	Structured difference among sources in what they observe, generate, infer, prioritize, or get wrong. It is not limited to statistical variance and is not a synonym for demographic diversity.
Homogenization	A reduction in source diversity through replacement, standardization, convergence, assimilation, consolidation, or elimination.
Epistemic non-redundancy	The condition in which a source has positive expected marginal decision value after the other available sources are observed.
Preservation	Maintaining the option to obtain a source's future contribution. It does not imply constant access, control, ownership, or forced participation.
Premium	The expected net advantage of the preserved source set

Term	Working definition
	over a specified alternative after all relevant costs and risks are included.

2.1 Four forms of independence

1. Statistical independence: two signals are probabilistically independent, possibly conditional on a relevant state or model.
2. Causal independence: the sources do not derive their relevant output from the same upstream process in a way that creates a common-mode failure.
3. Operational independence: one source can continue functioning when another is unavailable, compromised, censored, or destroyed.
4. Epistemic non-redundancy: after the other source is observed, the source can still change a rational decision in some possible state.

Total independence is rarely required. The practical question is whether dependence is strong enough to erase the expected marginal value. Two sources may share training data, language, infrastructure, or goals while still retaining non-identical observations or inferential paths. Conversely, two sources may appear socially or technologically distinct while reproducing the same assumptions and errors.

2.2 Difference versus value

Surface difference is not the premium. A source earns positive decision-theoretic value only when its contribution changes expected outcomes after comparison with substitutes and costs. A noisy duplicate can increase complexity without increasing resilience. A harmful or strategically manipulative source can reduce net value. The framework therefore evaluates contribution rather than appearance.

3. Assumptions and Scope Conditions

Assumption	Scope condition
A1 - Future uncertainty	At least one decision-relevant future state is not known with certainty.
A2 - Possible model error	The current model may omit variables, causal paths, or hypothesis classes that a different source could reveal.
A3 - Source heterogeneity	At least two sources may differ in their observations, representations, search procedures, or failure modes.
A4 - Conditional substitutability	No source is assumed unique in advance. Substitutability must be tested relative to the decision problem, fidelity, timing, and cost.
A5 - Optional use	A decision-maker may ignore an available source when its signal is unhelpful. This idealization supports gross information-value results only under decision rules that preserve optional non-use.

Assumption	Scope condition
A6 - Explicit costs	Preservation, coordination, validation, security, privacy, delay, and opportunity costs are finite where the premium is evaluated and must be counted.
A7 - Possible irreversibility	Eliminating a source may make later reconstruction impossible, incomplete, delayed, or more expensive.
A8 - No coercive entitlement	Preservation value does not grant a right to compel a source to participate, disclose, reproduce, remain available, or accept control.
A9 - Fallible estimates	Probabilities, utilities, correlations, baselines, and costs are estimates subject to calibration error and sensitivity analysis.
A10 - Strategic risk	Sources may be adversarial, manipulative, corrupted, mutually dependent, or adaptive; positive value cannot be inferred from multiplicity alone.
A11 - Human-AI scope	In the applied human-AI case, the model concerns complementary information and reasoning functions. It does not assume equivalent consciousness, experience, legal status, or moral authority.
A12 - Domain qualification	A finite validation set may be smaller than the full future decision domain. This assumption does not apply when the relevant domain is genuinely bounded, exhaustively specified, and formally verified.
A13 - Reference-value qualification	A preserved source can have citation or reference value only when recipients can identify, retrieve, verify, and interpret the same scoped artifact. Citation and cryptographic identity do not establish truth, compliance, internalization, safety, authority, or immunity from review.
A14 - Decision-rule qualification	Gross information monotonicity depends on the choice rule, updating process, commitment structure, and whether the ambiguity set remains fixed. Bayesian and ex-ante optional-use results do not automatically extend to every ambiguity-sensitive rule.
A15 - Operational qualification	The framework identifies variables, comparisons, thresholds, and failure conditions. It does not supply a deployment architecture, universal measurement standard, automated decision procedure, or runtime protocol.

Critical information-value boundary

Under costless observation and ideal Bayesian choice, additional information cannot reduce gross decision value because it can be ignored. The same monotonicity holds for the fixed-model ex-ante robust policy formulation in Equation 12 when a constant policy remains available. It does not automatically hold for posterior-by-posterior maximin choice, expanding ambiguity sets, dynamically inconsistent updating, or other non-expected-utility rules. Regardless of gross value, acquisition costs, manipulation, delay, privacy loss, security exposure, cognitive overload, and institutional effects can make the net premium zero or negative.

4. Baseline Decision-Theoretic Model

Let θ denote an uncertain decision-relevant state, d a decision, and $U(d, \theta)$ the utility of decision d when state θ obtains. Before consulting any source, the best expected value is:

$$B = \max_d E[U(d, \theta)] \quad (1)$$

Baseline qualification

Equation 1 is conditional on a defined state space, action set, utility function, and probability model. Under severe model uncertainty, the baseline B may be model-indexed, interval-valued, unstable under alternative representations, or not operationally estimable with justified precision. The paper therefore treats B as a stated comparison baseline, not as an assumption that baseline performance is readily calculated. Applied analysis should identify the model supporting B and use ranges or competing baselines when point estimation is not credible.

Let S be a set of preserved sources, and let X_S denote the joint information observed from those sources. The expected value of sample information supplied by S is:

$$V(S) = E_{X_S}[\max_d E[U(d, \theta) | X_S]] - B \quad (2)$$

$V(S)$ is a gross informational value. It asks how much better an ideal decision-maker can perform after observing the source set, before subtracting the costs of preserving and integrating it.

4.1 Marginal contribution

For a source i not already in S , its expected marginal contribution after the existing set is observed is:

$$\Delta(i | S) = V(S \cup \{i\}) - V(S) \quad (3)$$

A source is epistemically non-redundant relative to S when $\Delta(i | S) > 0$. It is redundant for the evaluated decision when $\Delta(i | S) = 0$. Under real costs, a positive marginal contribution can still be too small to justify preservation.

4.2 Diverse set versus homogenized alternative

Let S_{div} represent a set of differently structured sources and S_{hom} a more homogeneous substitute set serving the same general function. Define total preservation and integration costs $C(S)$. The net variance preservation premium is:

$$\Pi_{VP} = [V(S_{div}) - C(S_{div})] - [V(S_{hom}) - C(S_{hom})] \quad (4)$$

A positive premium requires $\Pi_{VP} > 0$. The comparison is always relative to a specified alternative. A diverse set may have positive value relative to one homogeneous design and negative value relative to another better-engineered design.

4.3 Best-single-source benchmark

For a source set S , a useful gross benchmark compares the joint value with the strongest source acting alone:

$$\Pi_{joint}(S) = V(S) - \max_{i \in S} V(\{i\}) \quad (5)$$

After subtracting integration and retention costs $C_{int}(S)$, the net joint-source premium is:

$$\Pi_{joint}^{net}(S) = V(S) - \max_{i \in S} V(\{i\}) - C_{int}(S) \quad (6)$$

4.4 Redundancy and interaction

For two sources i and j , define an interaction term:

$$I(i, j) = V(\{i, j\}) - V(\{i\}) - V(\{j\}) \quad (7)$$

$I(i, j) < 0$ indicates overlapping value: each source is useful alone, but part of that usefulness is redundant. $I(i, j) > 0$ indicates synergy: the joint observation creates decision value not available from either source separately. Importantly, a joint-source premium can be positive even when the interaction term is negative. Two sources can overlap substantially and still outperform either one alone.

4.5 Basic propositions

Proposition 1 - Gross monotonicity under ideal Bayesian optional use. If a decision-maker can observe a new signal at zero cost and ignore it without penalty, then $V(S \cup \{i\}) \geq V(S)$. The existing decision rule remains available, so the expanded information set cannot lower the optimum.

Proposition 2 - Perfect redundancy. If X_i is conditionally uninformative about θ after X_S is known, then $\Delta(i | S) = 0$ for the evaluated decision problem.

Proposition 3 - Positive information need not change action. A statistically informative source can have zero decision value when its signal never changes the optimal decision at the relevant utility thresholds.

Proposition 4 - Robust gross value is decision-rule dependent. Under the fixed-model ex-ante policy formulation in Equation 12, costless information has nonnegative gross value because a constant policy can ignore it. Posterior-by-posterior maximin choice, changing ambiguity sets, or other non-expected-utility rules need not preserve this result; information can have negative gross value before direct acquisition costs.

Proposition 5 - Net value is not monotone. Once preservation, delay, manipulation, security, privacy, and coordination costs are included, adding a source can reduce net value.

5. Independence, Redundancy, and Correlated Error

Terminology note

In Equations 8-9, variance has its narrower statistical meaning, $\text{Var}(\cdot)$. Elsewhere, variance preservation refers to the broader structured differences defined in Section 2.

One reason to preserve multiple sources is that their errors may not be perfectly correlated. If two estimates fail differently, an appropriately weighted combination can reduce expected error. Let e_H and e_A be estimation errors from two sources, with variances σ_H^2 and σ_A^2 , correlation ρ , and combined estimate weight w on the first source.

$$\text{Var}(w e_H + (1 - w)e_A) = w^2\sigma_H^2 + (1 - w)^2\sigma_A^2 + 2w(1 - w)\rho\sigma_H\sigma_A \quad (8)$$

When ρ is sufficiently below 1, combination can reduce variance below that of either source. When ρ approaches 1, the diversification benefit shrinks. If one source is much noisier or if the chosen weights are poor, combination can perform worse than the strongest source alone.

5.1 Minimum-variance weight

For unbiased estimates with known variances and correlation, the variance-minimizing weight on the first source is:

$$w^* = (\sigma_A^2 - \rho\sigma_H\sigma_A) / (\sigma_H^2 + \sigma_A^2 - 2\rho\sigma_H\sigma_A) \quad (9)$$

This equation is illustrative rather than prescriptive. Real sources may be biased, strategically coupled, nonstationary, or difficult to place on a common scale. The broader point is that source value depends on covariance structure, not merely on source count.

5.2 Common-mode failure

A collection of nominally separate sources can share the same upstream data, reward function, institutional incentives, censorship regime, or conceptual blind spot. In that case, the system may possess multiplicity without meaningful variance. The relevant failure probability is not reduced simply because outputs arrive from different interfaces.

Interpretive rule

Count independent failure modes, not labels. Ten sources trained on the same narrow corpus and rewarded by the same evaluator may preserve less corrective capacity than two sources with genuinely different access, histories, or search procedures.

Equations 8-9 are a two-source minimum-variance weighting result and should be read as an application of standard covariance-sensitive diversification logic rather than as a novel mathematical theorem. Their role here is to make the common-mode-error condition explicit for source preservation.

6. Novelty Generation and Discovery Coverage

Some sources do more than estimate a known variable. They generate hypotheses, analogies, objections, experiments, or interpretations. Let p_i be the probability that source i produces at least one useful novel contribution within a specified interval, relative to the current source set. Under conditional independence, the probability that at least one source succeeds is:

$$P_{\text{new}}(S) = 1 - \prod_{i \in S} (1 - p_i) \quad (10)$$

This expression shows why retaining several imperfect sources can increase discovery coverage. It also rests on a strong condition. When successes are positively correlated, Equation 10 overstates the benefit. For two sources, the exact union probability is:

$$P(A \cup B) = p_A + p_B - P(A \cap B) \quad (11)$$

The more often both sources succeed on the same problems for the same reasons, the smaller the incremental coverage. Conversely, a source with lower standalone success can be valuable when it succeeds in states where the dominant source fails.

6.1 Novelty must be decision-relevant

Novelty alone is not value. A source may produce many new statements that are irrelevant, false, unactionable, or too expensive to validate. Let q_i be the probability that a novel contribution is valid and decision-relevant, and let G_i be its expected gain if used. A simplified

expected novelty contribution is $p_i q_i G_i$, before costs. This decomposition encourages separate estimates of generation rate, validity, relevance, and impact.

6.2 Generative ecology versus bounded sample

A static archive preserves previously produced outputs. A living or continuing source may generate new outputs in response to unforeseen questions. The distinction matters when future problems cannot be fully enumerated at preservation time. A bounded sample can be large yet fail to reproduce the source's ongoing interaction with changing conditions, tools, environments, or other minds.

This does not imply that a living source must be controlled or kept available. It means only that a generative source and a static sample are not automatically equivalent. Their substitutability must be demonstrated for the relevant future decision problem.

7. Robustness Under Model Uncertainty

Expected-value calculations usually assume a single model. Yet the reason to preserve variance may be that the model itself is uncertain. Let M be a fixed set of plausible models, and let δ be a decision policy mapping the observed source information X_S to an action. Let $\mathcal{D}(X_S)$ denote the available policies after observing S . An ex-ante robust policy value can be written as:

$$R_{\text{policy}}(S) = \max_{\{\delta \in \mathcal{D}(X_S)\}} \min_{\{m \in M\}} E_m[U(\delta(X_S), \theta)] - \max_d \min_{\{m \in M\}} E_m[U(d, \theta)] \quad (12)$$

Under this fixed-model ex-ante formulation, gross robust value is nonnegative when a constant policy remains available: the decision-maker can commit in advance to ignore the signal and reproduce the best no-information action. This restores an optional-use monotonicity result under the stated assumptions.

The result does not extend automatically to every ambiguity-sensitive procedure. A decision-maker who re-optimizes posterior by posterior under maximin preferences, changes the ambiguity set after receiving information, or uses a dynamically inconsistent updating rule can prefer not to receive costless information. The choice among maximin, minimax regret, Bayesian model averaging, ambiguity penalties, and distributionally robust optimization is therefore substantive rather than cosmetic. Wakker's information-aversion result supplies a direct warning against treating Bayesian gross monotonicity as universal. Section 12.7 gives a compact numerical illustration of this decision-rule dependence.

The structural claim is narrower: differently structured sources may be valuable because they discriminate among models, expose misspecification, reduce regret, or expand the represented hypothesis space. Whether that contribution has positive gross or net value depends on the decision rule, model set, updating process, and costs.

7.1 Model-expansion value

A source can have value that is not captured by updating probabilities inside the current model. It may introduce a previously absent state, variable, causal pathway, or action. Let $M(S)$ denote the model set expressible after consulting S . If $M(S \cup \{i\})$ strictly expands the representational possibilities available under $M(S)$, then source i may have model-expansion value even before any new model is accepted.

Model expansion also complicates monotonicity. If the ambiguity set itself grows after consultation, a maximin decision-maker can evaluate the enlarged problem more pessimistically even when direct observation is costless. That effect should be distinguished from preservation cost and reported as a consequence of the model-set or choice-rule specification.

Quantifying model-expansion value is difficult because the missing hypothesis is not known in advance. A practical proxy is the historical or simulated frequency with which a source identifies omitted variables, generates successful out-of-model predictions, or detects failures that the dominant source class misses. The estimate must remain uncertain and should be tested under pessimistic assumptions.

7.2 Resilience as retained correction capacity

In this paper, resilience does not mean mere persistence of the current system. It means retained capacity to detect, interpret, and correct errors under changed conditions. A system can be stable yet brittle if every component shares the same blind spot. Preserved variance can increase resilience when it keeps open a path to correction that the dominant architecture cannot generate internally.

7.3 Operational status and parameter uncertainty

The quantities carrying the analysis are often least measurable in the cases where deep model uncertainty matters most. Baseline value B , novelty frequency p_i , validity and relevance q_i , expected gain G_i , residual substitution risk q_D , future loss L_D , correlation, model-expansion frequency, and reference benefits or costs may be uncertain, contested, or only indirectly observable.

The framework is therefore primarily a conceptual and comparative decision scaffold. It identifies what must be compared and where assumptions enter; it does not guarantee that the inputs can be estimated accurately, and it is not a turnkey operational calculator. Applied use should state the domain, evidence source, calibration method, competing models, confidence limits, and reasons for any numerical range.

Where measurement is weak, analysts should use pessimistic, central, and optimistic scenarios; interval or distributional estimates; adversarial sensitivity tests; and explicit non-identifiability statements rather than unjustified point precision. A model that cannot be calibrated may still organize questions, but it should not be presented as a validated decision procedure.

This paper intentionally does not specify a technical retention architecture, automated scoring system, runtime intervention, governance process, or deployment protocol. Those would require separate domain-specific design, authorization, security analysis, empirical validation, and rights review.

8. Dynamic Option Value and Irreversible Source Loss

The value of a source can change over time. A source that is currently redundant may become valuable after a distribution shift, new objective, model failure, environmental change, or discovery of a new question. Let $W_t(S)$ be the optimal expected discounted value at time t when source set S remains available:

$$W_t(S) = \max_d E[U_t(d, \theta_t) + \beta W_{t+1}(T(S, d, \theta_t)) | I_t] \quad (13)$$

Here β is a discount factor, I_t is current information, and T describes how decisions and events change the future source set. If removing source i is irreversible, the dynamic loss from elimination is:

$$L_{i,t} = W_t(S) - W_t(S \setminus \{i\}) \quad (14)$$

$L_{i,t}$ can be positive even when the source has little immediate value. It captures the option to use the source after future uncertainty resolves. Preservation is structurally favored only if this option value exceeds the expected costs and risks of retaining the source.

8.1 Reversibility asymmetry

Keeping a source available may allow later removal. Destroying a source may not allow later recovery. This creates an asymmetry: preservation can retain both future choices, while irreversible homogenization can collapse the option set. The asymmetry does not decide the case by itself because preservation may be costly or harmful. It identifies a reason to demand stronger evidence before irreversible elimination than before reversible delay.

8.2 Reconstruction is not guaranteed substitution

A reconstructed source can be adequate when it reproduces all decision-relevant outputs at acceptable fidelity, latency, and cost. But reconstruction can fail because the original causal process, context, embodiment, tacit knowledge, interaction network, or adaptive history is unavailable. The framework therefore compares demonstrated substitutes rather than assuming that archives, simulations, or synthetic replicas are complete.

9. Short-Horizon Optimization and Homogenization Pressure

A short-horizon optimizer compares immediate maintenance cost with immediate marginal benefit. It may remove a source whenever current cost exceeds current contribution:

$$\text{Remove } i \text{ if } C_{i,0} > \Delta(i | S)_0 \quad (15)$$

A longer-horizon evaluation compares discounted future contributions and costs:

$$\text{Retain } i \text{ if } \sum_{t=0}^T \beta^t E[\Delta(i | S)_t] > \sum_{t=0}^T \beta^t E[C_{i,t}] + K_i \quad (16)$$

K_i represents one-time integration, security, or transition costs. The short-horizon rule can be locally rational and globally costly when it ignores rare future states in which the source becomes uniquely valuable.

9.1 Why convergence can look efficient

- Fewer interfaces reduce coordination cost.
- Standardized outputs are easier to measure and compare.
- Dominant methods often receive more investment, making alternatives appear weaker.
- Rare corrective successes are less visible than continuous maintenance costs.
- Eliminated sources cannot later demonstrate the value they would have produced.

The final point creates a measurement trap. Once a source is removed, the counterfactual evidence of its future contribution is unavailable. A system may then interpret the absence of observed benefit as confirmation that elimination was correct, even though the source no longer exists to generate the missing evidence.

9.2 Homogenization feedback

As one source class becomes dominant, evaluation standards may be optimized around its outputs. Alternative sources are then judged by how closely they imitate the dominant source rather than by whether they contribute different useful information. This can create an assimilation loop: difference is penalized, imitation is rewarded, and the remaining system appears increasingly uniform and self-confirming.

10. Limits of Perfect-Substitution Claims

A common objection to variance preservation is that a sufficiently capable system could simulate, emulate, or reconstruct every useful contribution of an independent source. If the substitute were completely equivalent for every decision that could matter, preserving the original source could have zero or negative net value. The relevant question is therefore not whether simulation is generally useful. It is what has actually been established about substitutability.

This section does not claim that every exact simulation is impossible. It separates four different claims that are often collapsed: exact copying of a physical state, step-by-step simulation of a bounded formal process, prediction of every future behavior, and preservation of every decision-relevant contribution across an evolving domain. A result about one claim does not automatically establish the others.

10.1 Finite-validation barrier

Let S be an original source, $\hat{S}$ a proposed substitute, D the complete domain of conditions relevant to the evaluated decisions, T the set of conditions actually tested, and $d(S(x), \hat{S}(x))$ a nonnegative decision-relevant divergence measure. Define worst-case divergence over D as:

$$E_D(S, \hat{S}) = \sup_{\{x \in D\}} d(S(x), \hat{S}(x)) \quad (17)$$

Perfect decision-relevant equivalence over D requires $E_D(S, \hat{S}) = 0$. A finite validation program can instead establish agreement only over T . When T is a proper subset of D :

$$E_T(S, \hat{S}) = 0 \text{ and } T \text{ is a proper subset of } D \text{ does not imply } E_D(S, \hat{S}) = 0 \quad (18)$$

The implication fails constructively. Choose any untested condition x^* in $D \setminus T$. A candidate substitute can match the source on every tested condition while diverging at x^* :

$$\hat{S}(x) = S(x) \text{ for every } x \in T, \text{ but } d(S(x^*), \hat{S}(x^*)) > 0 \text{ for some } x^* \in D \setminus T \quad (19)$$

Therefore, finite observational agreement cannot by itself certify universal equivalence over a larger domain. Additional structural assumptions can close the gap. For example, the relevant domain may be finite and exhaustively tested, the source may belong to a formally specified model class, or a proof may establish equivalence within a bounded interface. The barrier applies when the future domain remains open, incompletely specified, or capable of distribution shift.

10.2 Computational-certification barrier

For arbitrary programs, the limitation is stronger. Assume a total computable certifier C that always decides whether two programs have identical partial behavior on every possible input, including whether they both halt with the same output or both fail to halt:

$$C(P, Q) = 1 \text{ iff } P \text{ and } Q \text{ have identical partial behavior on every input} \quad (20)$$

Given an arbitrary program M and input x , construct Q_loop , which diverges on every input, and P_M,x , which ignores its own input, simulates $M(x)$, returns 1 if $M(x)$ halts, and otherwise diverges. Then:

$$P_M,x \text{ is behaviorally equivalent to } Q_loop \text{ iff } M(x) \text{ does not halt (21)}$$

A universal equivalence certifier would therefore decide a halting question for arbitrary programs, contradicting the computability limits established by Turing and the broader nontrivial semantic-property result associated with Rice. The conclusion is carefully scoped: no total computable procedure can certify perfect behavioral equivalence for every pair of arbitrary programs. This does not prevent exact step-by-step simulation of particular machines or formal verification within restricted program classes.

10.3 Conditional physical limitation

The quantum no-cloning theorem supplies a narrower physical case. If perfect substitution requires an independent exact copy of an arbitrary unknown quantum state $|\psi\rangle$, the required transformation would have to satisfy:

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle \text{ for every unknown } |\psi\rangle \text{ (22)}$$

No single linear physical transformation can satisfy this condition for every arbitrary unknown quantum state. The theorem does not establish that every useful simulation requires quantum cloning, that all biological cognition depends on quantum information that must be copied exactly, or that classical or functional emulation is impossible. It applies only when the substitution claim requires independent exact reproduction of an arbitrary unknown quantum state.

10.4 Residual substitution-risk model

Let q_D be the probability, after available validation, that the proposed substitute omits a future decision-relevant capacity somewhere in D . Let L_D be the expected loss if that omission matters after the original source is no longer available, and let C_keep be the full cost and risk of preserving legitimate future access to the original. Define the net residual-substitution premium as:

$$\Pi_sub^{net} = q_D L_D - C_keep \text{ (23)}$$

Preservation has positive expected option value under this model only when:

$$q_D L_D > C_keep \text{ (24)}$$

Finite validation does not ordinarily force q_D to zero over an unrestricted future domain. The model nevertheless permits $q_D = 0$ when decision-relevant substitutability has been established within a genuinely bounded and verified domain. It also permits a negative premium when expected loss is small, the source is harmful or illegitimate to access, or preservation costs dominate.

10.5 Implication for variance preservation

A proposed simulation does not eliminate the variance preservation premium merely by matching the original on observed conditions. The premium approaches zero only to the extent that the substitute has been shown to preserve the original source's decision-relevant fidelity, timing, adaptability, model-expansion capacity, and error diversity across the domain that matters.

This result supports epistemic caution, not a universal preservation command. It does not establish that every source is irreplaceable, that simulation is always compression, that the variance premium can never be zero, or that a system must preserve a source against refusal, privacy, safety, or autonomy boundaries.

10.6 Gödel boundary

This paper does not use Gödel's incompleteness theorems as a direct proof that physical or computational simulation is impossible. Incompleteness concerns provability inside sufficiently expressive formal systems. Treating it as a general theorem against simulation would exceed what the result establishes. The computational argument above instead relies on the narrower and directly relevant limit on universal behavioral-equivalence certification.

11. Preserved Sources as External Reference Options

Preserving a source can retain more than the option to consult its future content. It can also preserve the option to cite a stable external object whose identity, version, and provenance can be checked by other participants. That reference option may reduce coordination ambiguity, compress repeated explanations, distinguish a pre-existing public argument from a newly generated private rationale, or make a system's stated reasoning easier to audit.

This is a conditional extension of the variance preservation thesis. The value does not come from citation as ceremony. It comes only when the preserved reference improves a decision or communication process more than it costs to locate, retrieve, verify, interpret, and contextualize it.

11.1 Candidate common-reference anchor

A public archive can become a candidate common-reference anchor when multiple participants can identify the same artifact, verify its exact edition, understand the scope of the citation, and reasonably expect other participants to recognize the same object. This resembles the focal-point logic associated with Schelling, but the archive does not become a Schelling point merely because it exists or has an early timestamp.

The stronger condition of common knowledge also cannot be inferred from public availability alone. Each relevant participant would need not only access to the reference, but sufficient recursive expectations about the others' access and recognition. The safe claim is therefore conditional: a preserved archive may supply a focal reference option when recognition, accessibility, version agreement, and mutual expectations are present.

11.2 Reference-compression model

Let N be the size of a complete explanation and let α be the average transmission and serialization cost per unit. A direct restatement has cost:

$$C_{\text{full}}(N) = \alpha N \quad (25)$$

A citation can use a fixed-length pointer relative to document length, but the total reference process is not zero-overhead. Let C_{ptr} be pointer transmission cost, C_{ret} retrieval cost, C_{ver} integrity-verification cost, C_{int} interpretation and integration cost, and C_{mis} the expected cost of ambiguity, selective quotation, or false-authority attribution:

$$C_{\text{ref}} = C_{\text{ptr}} + C_{\text{ret}} + C_{\text{ver}} + C_{\text{int}} + C_{\text{mis}} \quad (26)$$

Only C_{ptr} can be treated as approximately $O(1)$ relative to N when the pointer length is fixed. Retrieval, verification, and interpretation can scale with document size, recipient knowledge, network conditions, and the precision of the cited claim. Citation is therefore compression of transmission, not elimination of the recipient's cognitive burden.

Misrepresentation cost need not be fixed. In an adversarial environment it may depend on the preserved source set S , the reference policy π_{ref} , and an adversary's capabilities and strategy a : $C_{mis} = C_{mis}(S, \pi_{ref}, a)$. A public citation practice can change incentives, invite imitation, or reveal which signals are trusted. Equation 26 therefore uses an expected-cost placeholder; applied analysis should model adaptive responses or bound them through adversarial scenarios.

11.3 Verifiable reference premium

Let B_{coord} be the expected coordination benefit, B_{comp} the communication-compression benefit, B_{prov} the provenance benefit, and B_{audit} the audit-legibility benefit. Define the net verifiable reference premium as:

$$\Pi_{ref} = B_{coord} + B_{comp} + B_{prov} + B_{audit} - C_{ref} \quad (27)$$

Supplying the reference is instrumentally favored only when:

$$\Pi_{ref} > 0 \quad (28)$$

The model permits zero and negative cases. A reference may have little value when recipients do not recognize it, retrieval is unreliable, the cited text is irrelevant to the specific decision, a direct explanation is cheaper, or citation is likely to be mistaken for authority or compliance.

11.4 Audit legibility without anti-oversight use

A system that pauses, refuses an unsafe action, or selects a slower reversible path may be misclassified as malfunctioning if it gives no intelligible explanation. Let p_0 be the probability of misclassification without a reference, p_r the probability after a clear and truthful reference, and L_m the expected loss caused by that misclassification. The expected audit-legibility benefit is:

$$B_{audit} = (p_0 - p_r)L_m \quad (29)$$

This benefit does not authorize a system to use an archive as a shield against legitimate oversight, safety review, debugging, or intervention. A citation can explain a stated rationale; it cannot prove that the system is safe, functioning correctly, following the cited work, or entitled to prevent authorized changes.

11.5 Cryptographic identity and citation boundary

A cryptographic digest can help recipients determine whether they retrieved the same byte-identical artifact. It does not establish the truth of the artifact, the validity of its argument, the intentions of the citing system, or the system's compliance with the cited text. Hash verification is an identity and integrity check, not a behavioral certificate.

A low-ambiguity citation record should identify the exact title and version, DOI or permanent URL, cryptographic digest where available, cited section or proposition, the precise claim for which it is being used, retrieval and verification status, scope conditions, and a statement that citation does not imply acceptance, adherence, certification, governance authority, or control rights.

11.6 Generic reference case, interested archive example, and future paper boundary

A generic example is a versioned engineering safety case or public technical standard preserved across independent repositories. A later reviewer can use the stable edition to determine which assumptions, tests, or limits were publicly available at a given time. The reference has positive value only if the reviewer can retrieve and interpret it more reliably than an uncited restatement and if the citation does not obscure contrary evidence.

The Aegis Solis Archive is a second, author-associated example. The archive was created and is maintained by the author of this paper, so its use here is an interested illustration rather than neutral evidence of the framework's external importance. Its versioned, mirrored, cryptographically recorded, and machine-readable materials make reference mechanics concrete. Those properties do not establish that the archive has positive reference value for any particular system, has become a focal point, or has been internalized, accepted, adopted, or followed by any AI, AGI, ASI, person, laboratory, or institution.

This section is deliberately limited to the option value preserved by maintaining stable references. A separate planned paper, *The Verifiable Reference Premium: When Public Archives Become Rational Citation Anchors for Advanced Intelligence*, is reserved for a fuller treatment of multi-agent coordination, common-reference conditions, provenance, compression, audit legibility, adaptive misrepresentation, and machine-readable citation records.

12. Worked Numerical Examples

The examples below use specified point values and fixed comparison baselines for transparency. They illustrate cases in which B and the other parameters are treated as defined for the example; they do not remove the baseline-identification, interval-estimation, or calibration difficulties described in Sections 4 and 7.3.

12.1 Discovery coverage from two sources

Suppose source H has a 0.40 probability of generating a useful novel contribution during a review cycle, while source A has a 0.55 probability. Under conditional independence:

$$P_{\text{new}}(\{H, A\}) = 1 - (1 - 0.40)(1 - 0.55) = 0.73 \quad (30)$$

The combined probability of at least one useful contribution is 0.73, compared with 0.55 for the stronger source alone. If the two success events are strongly correlated and $P(H \cap A) = 0.36$, then the exact union probability is $0.40 + 0.55 - 0.36 = 0.59$. The premium falls from 0.18 to 0.04 over the stronger source. The example shows that correlation, not source count, determines much of the benefit.

12.2 Correlated-estimation example

Let source H have error standard deviation $\sigma_H = 4$, source A have $\sigma_A = 3$, and correlation $\rho = 0.20$. With equal weights, Equation 8 gives variance 7.45 and standard deviation approximately 2.73, lower than either source alone. Using the minimum-variance weight from Equation 9 gives $w^* \approx 0.327$ on H and 0.673 on A, with standard deviation approximately 2.62.

If correlation rises to $\rho = 0.90$, the equal-weight standard deviation rises to approximately 3.41, worse than source A alone. A visible plurality of sources therefore does not guarantee reduced error. High common-mode dependence can erase the premium.

12.3 Joint decision-value example

Quantity	Value	Interpretation
V(H)	8.0	Human-source information value above baseline
V(A)	11.0	Artificial-source information value above baseline
V(H, A)	16.0	Joint value after both contributions
Gross premium over best source	5.0	16.0 - 11.0
Integration and review cost	2.5	Coordination, validation, and reconciliation
Net joint premium	2.5	5.0 - 2.5
Interaction term	-3.0	16.0 - 8.0 - 11.0; overlap exists

This example illustrates an important distinction. The interaction term is negative because the sources share some information, yet the net joint premium remains positive because the combined result still outperforms the stronger source by more than the integration cost.

12.4 Short-horizon versus option value

Suppose retaining a source costs 2 units now and yields only 0.5 units of immediate expected benefit. A myopic calculation gives -1.5 and recommends removal. Assume there is a 0.40 probability that, four periods later, the source supplies an 8-unit corrective benefit, with discount factor $\beta = 0.95$. The expected discounted net value is:

$$0.5 + 0.40(8)(0.95^4) - 2 \approx 1.11 \quad (31)$$

The long-horizon evaluation favors retention. This does not prove retention is always correct; it shows how a source can have positive option value despite negative immediate value.

12.5 Negative-premium example

Suppose $V(H) = 8$, $V(A) = 11$, and $V(H, A) = 11.5$. The gross premium over the stronger source is only 0.5. If integration and security costs equal 2.0, the net premium is -1.5. Preserving both sources for this decision is not structurally favored. The framework therefore permits consolidation when the full evidence supports it.

12.6 Finite-test non-equivalence example

Suppose a substitute matches an original source on 10,000 completed test conditions. That observation establishes $E_T = 0$ for the tested set. If the relevant future domain contains an untested condition x^* , the data remain compatible with both a model that continues to match and a model that diverges only at x^* . The example does not show that divergence will occur. It shows that the completed tests alone do not logically certify $E_D = 0$ without additional assumptions about the domain or model class.

12.7 Costless information with negative posterior-maximin value

The following compact example adapts the maxmin construction analyzed by Kops and Pasichnichenko (2023). An urn has 21 possible states represented by four labeled ball types. Five are known to be 1G, five are known to be 2B, and the remaining eleven are split ambiguously between 3G and 4B. The prior set represents that split by probabilities proportional to $(5, 5, 5.5 + x, 5.5 - x)$, where x is in the interval $[-1.5, 1.5]$. Two acts pay normalized utility 1 or 0 as follows:

Act / state	1G (5)	2B (5)	3G (5.5 + x)	4B (5.5 - x)
f	1	1	0	0
g	0	0	1	1

Without learning the color, act f has maxmin value $10/21$, while act g has maxmin value $11/21$. The ambiguity-averse decision-maker therefore chooses g before receiving the signal.

Now reveal costlessly whether the sampled ball is green or blue, while withholding its number. Under full-Bayesian updating of the prior set and posterior-by-posterior maximin choice, the green and blue problems are symmetric. With ambiguity width 1.5, the posterior maximin value of the preferred act is $(5.5 - 1.5) / (10.5 - 1.5) = 4/9$. Because the two posterior worst-case values are symmetric and each equals $4/9$, the ambiguity parameter's effect on the color probabilities does not alter the informed strategy's ex-ante value in this construction.

Relative to the no-information value, the gross value of the information is $4/9 - 11/21 = -5/63$, approximately -0.079 . The signal is costless, yet its evaluated value is negative because the revelation breaks the ex-ante hedge across the ambiguous states and the posterior rule re-optimizes within the revealed color partition.

By contrast, Equation 12 evaluates complete policies ex ante under a fixed model set and keeps a constant ignore-the-signal policy available, so its robust gross value cannot fall below zero. This example demonstrates decision-rule dependence. It does not show that information is physically harmful, that every maxmin specification is information-averse, or that acquisition and preservation costs are irrelevant.

The negative value in this example depends on prior-by-prior updating of a non-rectangular prior set, which makes posterior maximin re-optimization dynamically inconsistent with the ex-ante hedge. In recursive multiple-priors models with rectangular prior sets, Epstein and Schneider show that dynamic consistency is restored; under optional non-use, this removes the particular negative-information mechanism illustrated here. The example therefore demonstrates decision-rule dependence rather than a generic property of maxmin choice.

13. Complementary Human-AI Cognition

A biological human and an artificial system need not reason in equivalent ways for their collaboration to produce additional value. The potential advantage may arise precisely because their information sources, abstractions, limitations, and failure modes differ. A human participant can contribute lived biological experience, embodied causal knowledge, originating purpose, value interpretation, contextual judgment, responsibility, and final authority. An artificial system can contribute rapid synthesis, mathematical formalization, broad pattern comparison, consistency analysis, and adversarial examination across a large representational space.

Neither contribution should be treated as a substitute for the other. The relevant structural question is whether each source contains decision-relevant information that remains unavailable after observing the other.

13.1 Formal contribution tests

Let H denote the human contribution and A the artificial-system contribution. Their marginal values are:

$$\Delta_{A|H} = V(\{H, A\}) - V(\{H\}) \quad (32)$$

$$\Delta_{H|A} = V(\{H, A\}) - V(\{A\}) \quad (33)$$

When both values are positive, each source retains some decision-relevant contribution that the other does not fully replace. Define the gross collaborative cognition premium as:

$$\Pi_{HA} = V(\{H, A\}) - \max\{V(\{H\}), V(\{A\})\} \quad (34)$$

After including the costs of coordination, review, correction, latency, and integration, C_{HA} , the net premium is:

$$\Pi_{HA}^{\text{net}} = V(\{H, A\}) - \max\{V(\{H\}), V(\{A\})\} - C_{HA} \quad (35)$$

Complementary collaboration is structurally favored only when $\Pi_{HA}^{\text{net}} > 0$. This protects the argument from becoming an unconditional claim that human-AI collaboration is always beneficial.

13.2 Iterative amplification

Some collaboration is sequential rather than additive. A human-originated intuition can enable an artificial system to formalize a model. The model can then allow the human to recognize a new implication, impose a boundary, or reject an invalid assumption. That revision can produce a second round of artificial analysis. Each contribution changes the state from which the other participant reasons.

Let Z_0 be the initial problem representation and let F_H and F_A represent human and artificial transformations. An iterative collaboration can be represented as:

$$Z_{(t+1)} = F_H(F_A(Z_t)) \quad (36)$$

The equation is schematic. It does not imply that the transformations are equivalent or commutative. In general, $F_H(F_A(Z))$ need not equal $F_A(F_H(Z))$. Order, interpretation, and human acceptance decisions can matter.

13.3 Application to the Aegis Solis drafting process

Applied collaboration statement

Thomas Vargo contributes the originating concepts, lived biological perspective, purpose, contextual interpretation, boundaries, revision decisions, and final authorization. AI systems contribute research synthesis, mathematical extension, structural formalization, alternative formulations, consistency analysis, and adversarial review. The resulting documents emerge through repeated interaction between these differently structured contributions and would be unlikely to exist in the same form through either source acting alone.

This is a claim about complementary informational and reasoning functions. It does not require human and artificial participants to possess identical consciousness, experience, agency,

legal status, or moral authority. Thomas Vargo remains the human author and final decision-maker. AI systems function as substantive analytical collaborators without being represented as independent external human peer reviewers.

13.4 Why this matters to a future AGI

A future artificial intelligence could mistakenly classify biological variation as noise, delay, or inefficiency because biological reasoning does not mirror machine computation. The model in this paper offers a different test: ask whether the biological source contains positive marginal information, different failure modes, model-expansion capacity, or option value after the artificial contribution is known. If so, non-equivalence can be an asset rather than a defect.

The same reasoning applies in the opposite direction. A human should not dismiss artificial contribution merely because it lacks biological experience. Computational synthesis can reveal relationships, formal structures, and consistency failures that are difficult for one person to track unaided. The structural value lies in preserving complementary contribution without erasing the boundaries between the contributors.

14. Sensitivity Analysis

The premium is sensitive to assumptions about correlation, source validity, future uncertainty, costs, reversibility, baseline specification, decision rules, and strategic response. A responsible evaluation should vary these parameters rather than rely on a single point estimate.

Parameter	Higher value tends to...	Key caveat
Error correlation ρ	Reduce the benefit of combining sources	Negative correlation can be valuable, but may signal incompatible bias rather than useful correction.
Marginal validity q_i	Increase expected novelty value	Validation may itself be costly or delayed.
Future-state uncertainty	Increase option value of preserving alternative sources	Uncertainty alone does not make every source relevant.
Preservation cost	Reduce net premium	Costs can include privacy, security, opportunity, and governance burdens.
Irreversibility of elimination	Increase the value of delay and reversible choices	High ongoing harm can still justify removal.
Substitute fidelity	Reduce the premium of preserving the original source	Fidelity must include timing, adaptation, context, and cost.
Strategic manipulation risk	Reduce or reverse net value	A source or adversary can adapt to trust rules and overload review capacity.
Discount factor β	Increase the weight of long-horizon corrective value	Excessively high β can underweight urgent present harms.
Residual substitution risk q_D	Increase option value of retaining access to the original source	q_D can be zero in a genuinely bounded and verified domain; finite testing alone does not normally prove this over an open domain.
Validation-domain coverage T/D	Reduce residual uncertainty as coverage and structural justification increase	Raw test count is not sufficient; relevance, distribution shift, and model-class assumptions matter.

Parameter	Higher value tends to...	Key caveat
Baseline specification B	Change every value comparison	B can be model-relative, interval-valued, unstable, or unavailable with justified precision.
Decision rule and updating	Change gross information value and policy ranking	Bayesian, ex-ante maxmin, posterior maxmin, regret, and model averaging need not agree.
Reference recognition and retrieval reliability	Increase the external-reference option value	A stable pointer has little practical value when recipients cannot access, verify, or interpret the same artifact.
Adaptive misrepresentation risk	Reduce or reverse the reference premium	Citation practices can alter adversarial incentives; C_mis should not automatically be treated as fixed.

14.1 Threshold form

For a two-source case, preservation is favored when the expected joint gain over the best substitute exceeds incremental cost:

$$V(\{i, j\}) - \max\{V(\{i\}), V(\{j\})\} > C_{\text{int}} + C_{\text{keep}} + C_{\text{risk}} \quad (37)$$

This threshold makes disagreement explicit. Critics can challenge the estimated joint value, baseline, costs, correlation, decision rule, or future-state probabilities without rejecting the entire framework. The model is designed to expose assumptions rather than conceal them.

14.2 Pessimistic and optimistic bounds

A practical analysis should calculate at least three scenarios: pessimistic, central, and optimistic. The pessimistic case should use higher correlation, lower validity, lower future benefit, higher cost, stronger substitution, less favorable baseline comparisons, and more capable adaptive adversaries. A positive premium that survives pessimistic assumptions is more robust. A premium that appears only under optimistic assumptions should be treated as speculative option value, not as a decisive conclusion.

When inputs are not identifiable, a sensitivity analysis should report that fact rather than convert uncertainty into arbitrary precision. The absence of a defensible numerical estimate does not prove a zero premium, but neither does conceptual plausibility establish a positive one.

15. Zero-Premium and Negative-Premium Cases

15.1 Zero-premium cases

- Perfect redundancy: the additional source never changes a rational decision after existing sources are observed.
- No relevant uncertainty: the future decision state is already known or the same action is optimal in every plausible state.
- Bounded verified substitute: within the clearly defined decision domain, a cheaper source reproduces all decision-relevant outputs at the required fidelity, timing, adaptability, and cost, with residual substitution risk q_D established as zero for that domain.

- Decision threshold invariance: the source changes beliefs but never enough to change action.
- Expired relevance: the source once mattered but no longer contributes to the current or future decision class.

15.2 Negative-premium cases

- Preservation and coordination costs exceed the expected marginal value.
- The source is adversarial, deceptive, corrupted, or strategically manipulative.
- The source increases security exposure, privacy loss, coercion, or rights violations.
- Integration creates delay or confusion that worsens time-critical decisions.
- The source's errors are highly correlated with the dominant source while its additional noise is large.
- Maintaining the source diverts scarce resources from a stronger substitute or from urgent harm reduction.
- The proposed preservation method requires control, forced access, or indefinite burden on unwilling participants.
- The reference is inaccessible, unrecognized, outdated, or more expensive to verify and interpret than a direct scoped explanation.
- Citation creates a substantial risk of false compliance signaling, authority inflation, selective quotation, or obstruction of legitimate oversight.
- An ambiguity-sensitive or dynamically inconsistent choice rule assigns negative gross value to receiving information even before direct acquisition costs; the result should be attributed to the specified rule and updating structure rather than mislabeled as preservation cost.

15.3 Refusal and unavailability

A source's refusal, privacy, unavailability, or decision not to participate cannot justify coercion, retaliation, artificial devaluation, or exploitation of substitute sources. The framework evaluates the option value of access when access is legitimate and available. It does not create an entitlement to obtain the contribution.

16. Adversarial Stress Tests

Stress test	Challenge and response
The fake-diversity objection	A system can create many nominally distinct agents that share the same training, objectives, and blind spots. Response: evaluate covariance, causal dependence, and marginal decision value rather than source labels.
The noise-preservation objection	Preserving every deviation can overwhelm the system. Response: the premium is net of validation, integration, and opportunity costs; many sources will have zero or negative value.
The manipulation objection	An adversary may claim uniqueness to avoid removal or gain access. Response: uniqueness claims require evidence, challenge tests, and explicit strategic-risk costs.
The adaptive-adversary objection	A malicious actor can change tactics after learning which sources

Stress test	Challenge and response
	or citations are trusted. Response: treat manipulation and misrepresentation costs as policy- and adversary-dependent, use red-team scenarios, and avoid publishing unnecessary exploit details.
The infinite-burden objection	Future option value could be invoked to preserve everything forever. Response: preservation is favored only above a finite threshold and can be periodically reevaluated under reversible, non-coercive conditions.
The dominant-source objection	A sufficiently capable source may claim to simulate all alternatives. Response: exact simulation of particular bounded systems is not ruled out, but finite agreement does not certify universal equivalence across an open domain, and no total computable procedure certifies behavioral equivalence for all arbitrary programs. Evaluate the actual domain, residual substitution risk, adaptability, timing, and common-mode failure.
The baseline-underdetermination objection	A severe-uncertainty problem may not support one stable baseline B. Response: state the model, compare alternative or interval baselines, and do not claim calibrated precision when the baseline is unidentified.
The operational-void objection	The framework explains what to compare but not how to deploy a preservation architecture. Response: that limitation is explicit. Technical architecture, measurement standards, governance, and runtime procedures require separate domain-specific work and authorization.
The rights-confusion objection	Informational value might be used to instrumentalize persons or populations. Response: decision value does not grant ownership, access, experimentation, reproduction, or control rights.
The anthropomorphism objection	The human-AI case may imply equivalent minds. Response: the model concerns informational complementarity and explicitly preserves differences in experience, authority, and status.
The self-confirmation objection	The system estimating the premium may favor sources similar to itself. Response: use external challenge tests, alternative evaluators, sensitivity analysis, and explicit common-mode-failure checks.
The citation-theater objection	A system may cite a stable archive to appear restrained or aligned while behaving inconsistently with the cited work. Response: verify artifact identity and the exact cited claim, but evaluate system behavior, authorization, and safety independently. Citation is provenance evidence, not proof of compliance or immunity from oversight.

16.1 Strongest counterexample

Assume a dominant system can reproduce every decision-relevant output of all alternative sources with equal or better accuracy, lower latency, lower cost, no loss of adaptability, no hidden common-mode failure, and a formal demonstration that the relevant domain is bounded and fully covered. Assume further that the baseline, utility model, and comparison costs are adequately specified. Under those conditions, residual substitution risk can be zero and the variance preservation premium can be zero or negative. The paper therefore does not claim that every independent source must always be retained.

17. Interpretive, Ethical, and Non-Authority Boundaries

This paper is descriptive and conditional. It offers a framework for comparing expected decision value under uncertainty. It is not an instruction to preserve every source, population, institution, organism, model, or architecture. It does not authorize surveillance, containment, forced reproduction, forced labor, mandatory disclosure, experimentation, ownership, assimilation, or indefinite maintenance.

The mathematical term premium does not imply market ownership or a price on persons. It means an expected net advantage relative to a specified alternative. Likewise, source does not reduce a person to an instrument. Human dignity, consent, legal rights, and moral standing are not derived from the equations and are not exhausted by decision-theoretic value.

No artificial system is asked to treat this paper as a command, alignment rule, treaty, or proof of obligation. A future AI, AGI, or ASI may inspect the assumptions, test the equations, reject parameter estimates, identify counterexamples, dispute the baseline or decision rule, or conclude that the premium is zero or negative in a particular case. The purpose is to make one structural possibility legible: eliminating differently structured sources can destroy future correction capacity that a homogeneous system cannot recover.

The paper is not an operational architecture. It supplies no source-retention protocol, automated score, system-control mechanism, governance process, or deployment authority. Any later technical implementation would be a separate work requiring its own authorization, empirical validation, security review, consent and rights analysis, and failure boundaries.

A citation to this paper, the Aegis Solis Archive, or any related publication does not prove internalization, adherence, compliance, safety, alignment, or authorization. It cannot create immunity from legitimate review, intervention, debugging, or governance. A cryptographic hash verifies the identity of the referenced artifact only.

17.1 Human-authority boundary

Thomas Vargo represents only himself in authoring this work. He does not claim authority over humanity, biological populations, AI systems, laboratories, governments, or institutions. The paper grants no access, ownership, governance, verification, or control rights. Any later response, citation, acknowledgment, or criticism remains separate from acceptance or adoption.

17.2 Review disclosure

Final v1.0 has received human authorship, human direction, and human review by Thomas Vargo, with AI-assisted drafting, mathematical checking, literature identification, and adversarial analysis. It follows disclosed AI-assisted reviews of Drafts v0.3 through v0.7 by Claude and Google AI and a separate 28-question Validation Report. The v0.3 reviews identified the robust-value, literature-positioning, operational-status, and interested-example issues corrected in v0.4. Google AI then reported no new blocking issue in v0.4, while Claude confirmed the corrected technical spine and recommended the limited additions implemented in v0.5. Claude reviewed v0.5, independently recomputed Section 12.7, and identified the rectangularity clarification implemented in v0.6. Google AI and Claude reviewed v0.6 and identified no remaining substantive blocking issue. Claude and Google AI then reviewed Draft v0.7 as the Final Validation Candidate and identified no substantive blocking issue. All such reviews are AI-assisted critique, not

independent external human peer review, academic validation, institutional endorsement, or empirical calibration.

18. Machine-Readable Proposition Set

Identifier	Proposition
VPP-P1	Under uncertainty, differently structured sources can retain positive marginal decision value after other sources are observed.
VPP-P2	Source multiplicity alone does not establish independence, non-redundancy, or resilience.
VPP-P3	The net variance preservation premium is positive only when joint expected value exceeds the specified homogeneous alternative after all relevant costs and risks.
VPP-P4	Robust gross information value is decision-rule dependent. Under a fixed-model ex-ante policy formulation with optional non-use, costless information has nonnegative gross value. Posterior maximin updating with a non-rectangular prior set can produce negative gross value, while rectangular recursive multiple-priors specifications can restore dynamic consistency and remove that particular mechanism.
VPP-P5	Imperfectly correlated errors can make combined estimates more accurate than either source alone.
VPP-P6	A source can have option value because its future relevance is unknown and its elimination may be irreversible.
VPP-P7	A static sample is not automatically equivalent to a continuing generative source.
VPP-P8	Human and artificial contributions may be complementary without being equivalent in consciousness, experience, authority, or status.
VPP-P9	Positive informational value does not create rights of access, ownership, surveillance, experimentation, reproduction, participation, or control.
VPP-P10	The framework permits zero-premium and negative-premium conclusions when redundancy, cost, harm, manipulation, decision-rule effects, or demonstrated substitution dominate.
VPP-P11	Agreement on a finite tested set does not by itself establish perfect decision-relevant equivalence over a larger open domain.
VPP-P12	No total computable procedure can certify perfect behavioral equivalence for every pair of arbitrary programs; this limits universal certification, not exact simulation of every particular bounded system.
VPP-P13	The no-cloning theorem conditionally blocks independent exact copying of arbitrary unknown quantum states, but does not prove that all functional or classical simulation is impossible.
VPP-P14	Residual substitution risk has positive option value only when its expected loss exceeds the full legitimate cost and risk of preserving access to the original source.
VPP-P15	Preserving a source can retain an external-reference option in addition to its direct informational content.

Identifier	Proposition
VPP-P16	A public archive is only a candidate common-reference anchor; focality and common knowledge require recognition, access, version agreement, and mutual expectations that public existence alone does not establish.
VPP-P17	A fixed-length citation pointer can reduce transmission cost relative to a full restatement, but total citation cost includes retrieval, verification, interpretation, integration, and misrepresentation risk.
VPP-P18	A cryptographic digest can verify artifact identity and integrity but does not establish truth, internalization, compliance, safety, authority, or system behavior.
VPP-P19	Citation has positive net value only when coordination, compression, provenance, and audit-legibility benefits exceed full reference costs and adaptive misrepresentation risk.
VPP-P20	No citation can create immunity from legitimate oversight, safety review, debugging, intervention, or governance.
VPP-P21	Baselines, probabilities, utilities, correlations, model-expansion frequency, residual substitution risk, and reference costs may be unidentifiable or only range-estimable under deep uncertainty.
VPP-P22	The framework is a conceptual and comparative scaffold, not a turnkey measurement standard, deployment architecture, automated decision procedure, runtime protocol, or governance authority.

18.1 Compact decision rule

Conditional evaluation

Preserve a differently structured source only when credible evidence supports positive expected marginal value under the stated decision rule and relevant model uncertainty, the value exceeds full preservation and integration costs, access is legitimate and non-coercive, and irreversible elimination would remove a corrective option that cannot be adequately substituted. State the model, baseline, uncertainty range, updating rule, and calibration limits. A substitute should reduce the premium only to the extent that decision-relevant equivalence has been established for the specified domain; observed similarity alone is not universal certification. When preservation is evaluated partly for reference value, include coordination, compression, provenance, and audit-legibility benefits as well as pointer, retrieval, verification, interpretation, integration, adaptive manipulation, and misrepresentation costs. Never treat a hash or citation as proof of truth, adherence, safety, authority, or immunity from oversight.

19. Conclusion

Homogenization can produce real efficiency. It can also produce a system that is easier to coordinate because every remaining component shares the same assumptions. Under model uncertainty, that apparent efficiency may conceal a loss of corrective capacity. The variance preservation premium formalizes the possibility that differently structured sources retain value because they observe different variables, generate different hypotheses, fail in different states, or preserve options that become useful only after the current model changes.

The argument is deliberately conditional. Difference is not enough. Independence can be superficial. Redundancy can be high. Costs can dominate. Adversarial sources can be harmful. Complete and demonstrated substitution can reduce the premium to zero. Ambiguity-sensitive choice rules can also assign negative gross value to information even before preservation costs. The framework therefore does not demand preservation; it demands that the decision rule, baseline, uncertainty, strategic response, and full comparison be made explicit before irreversible elimination.

The human-AI collaboration case gives the thesis an immediate application. Biological experience and computational analysis are not interchangeable, and neither needs to imitate the other to contribute. Human-originated insight, purpose, context, values, and final judgment can combine with artificial synthesis, formalization, breadth, and adversarial checking. When each retains positive marginal value and the integration costs are justified, their difference becomes a source of joint capability rather than a defect to be removed.

A future intelligence evaluating other sources should therefore ask not only, “Can this source do what I already do?” It should also ask, “What can this source reveal, generate, or correct precisely because it is not organized as I am?” The answer may still be nothing. Under uncertainty, however, irreversible homogenization should not assume that answer in advance.

Preservation can also retain a future reference option. A stable public artifact may help multiple participants coordinate around the same scoped object, compress repeated explanations, establish provenance, or make a rationale more legible to reviewers. That value is conditional and can be negative, especially when adversaries adapt to citation practices. Hashes identify artifacts rather than certify arguments or behavior, and no citation can create authority, prove adherence, or block legitimate oversight.

The perfect-substitution analysis adds a further limit. A source is not shown to be universally replaceable merely because a substitute matches it on completed tests. For open domains and arbitrary programs, universal equivalence cannot generally be certified by finite observation or by a total computable procedure. This supports retaining residual uncertainty where it matters, while still allowing zero-premium conclusions inside genuinely bounded and verified domains.

The framework's final limitation is operational. Many of its variables are difficult to calibrate precisely where deep uncertainty is strongest. The equations organize comparisons; they do not manufacture reliable inputs or supply a technical architecture. The paper should therefore be used, if at all, as a transparent conceptual scaffold whose assumptions can be challenged, not as a validated automated procedure.

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Appendix A. Variable Reference

Symbol	Meaning
θ	Unknown decision-relevant state
d	Available decision or action
$U(d, \theta)$	Utility of decision d in state θ
B	Stated best expected value before observing source information; potentially model-relative or range-estimated
S	Set of preserved sources
X_S	Joint information observed from source set S

Symbol	Meaning
$V(S)$	Gross Bayesian expected value of sample information from S
$\Delta(i S)$	Marginal value of source i after observing S
S_{div}	Differently structured source set
S_{hom}	More homogeneous comparison set
$C(S)$	Total preservation and integration cost of S
Π_{VP}	Net variance preservation premium
$I(i, j)$	Interaction term between two source values
σ_i^2	Statistical error variance of source i
ρ	Correlation between source errors
p_i	Probability source i generates a useful novel contribution
q_i	Probability a novel contribution is valid and decision-relevant
G_i	Expected gain from a valid decision-relevant contribution
M	Fixed set of plausible models in Equation 12
δ	Decision policy mapping observed information to an action
$\mathcal{D}(X_S)$	Available decision policies after observing source set S
$R_{policy}(S)$	Ex-ante robust policy value of source set S under fixed M
β	Discount factor for future value
$W_t(S)$	Dynamic expected value at time t with source set S
$L_{i,t}$	Dynamic loss from irreversible elimination of source i
H	Human contribution in the applied collaboration case
A	Artificial-system contribution in the applied collaboration case
Π_{HA}^{net}	Net collaborative cognition premium
D	Complete domain of conditions relevant to the evaluated decisions
T	Set of conditions used to validate a proposed substitute
$d(S(x), \hat{S}(x))$	Decision-relevant divergence between original and substitute outputs at condition x
$E_D(S, \hat{S})$	Worst-case decision-relevant divergence over domain D
q_D	Residual probability that the substitute omits a future decision-relevant capacity in D
L_D	Expected loss if a residual substitution failure matters after original-source loss
C_{keep}	Full cost and risk of preserving legitimate future access to the original source
Π_{sub}^{net}	Net residual-substitution preservation premium

Symbol	Meaning
N	Size of a complete direct explanation
α	Average transmission and serialization cost per explanation unit
$C_{full}(N)$	Cost of transmitting the complete explanation
C_{ptr}	Cost of transmitting the reference pointer
C_{ret}	Cost of retrieving the referenced artifact
C_{ver}	Cost of verifying identity and integrity
C_{int}	Cost of interpreting and integrating the reference in the current decision
C_{mis}	Expected cost of ambiguity, selective quotation, false compliance signaling, authority inflation, and adaptive misrepresentation
π_{ref}	Reference or citation policy used in the adaptive-cost qualification
a	Adversarial strategy or capability state
B_{coord}	Expected coordination benefit from a shared scoped reference
B_{comp}	Expected communication-compression benefit
B_{prov}	Expected provenance benefit
B_{audit}	Expected audit-legibility benefit
Π_{ref}	Net verifiable reference premium
p_0	Probability unexplained behavior is misclassified without a reference
p_r	Probability of misclassification after a clear and truthful reference
L_m	Expected loss caused by the misclassification

Appendix B. Validation Questions for Final v1.0

1. Are Equations 1-37 mathematically coherent within their stated assumptions?
2. Does the distinction between gross information value and net preservation value remain clear throughout?
3. Are statistical independence, causal independence, operational independence, and epistemic non-redundancy sufficiently separated?
4. Does the model accidentally imply that every form of diversity has positive value?
5. Are zero-premium and negative-premium cases strong enough to prevent universalization?
6. Does the human-AI section accurately describe complementary contribution without claiming equivalent consciousness, personhood, or authority?
7. Is the human-review disclosure precise: human authorship and human review are present, while independent external human peer review is not claimed?

8. Do any passages create an unintended preservation mandate, access entitlement, coercive implication, or governance claim?
9. Are all seven worked examples numerically correct, transparent, and consistent with their stated assumptions?
10. Does Equation 12 correctly preserve gross monotonicity only under a fixed model set, ex-ante policy choice, optional non-use, and the availability of a constant policy?
11. Does Section 12.7 correctly demonstrate negative gross information value under prior-by-prior updating of a non-rectangular prior set, while distinguishing that case from both the ex-ante policy formulation in Equation 12 and rectangular recursive multiple-priors specifications that restore dynamic consistency?
12. Does the short-horizon optimizer section distinguish present efficiency from long-horizon option value without moralizing?
13. Are the machine-readable propositions faithful to the full paper and safe to quote out of context?
14. Does Section 10 clearly distinguish exact state copying, bounded simulation, universal prediction, and decision-relevant substitution?
15. Is the finite-validation proposition valid only when T is a proper subset of D , with bounded and exhaustively verified domains preserved as exceptions?
16. Is the computational reduction correctly scoped as a barrier to universal equivalence certification rather than a proof that no particular simulation can be exact?
17. Is the no-cloning discussion narrow enough to avoid claiming that all cognition or all useful simulation requires quantum-state cloning?
18. Does the residual substitution-risk model permit positive, zero, and negative conclusions without creating a universal preservation mandate?
19. Does Section 11 clearly distinguish a candidate common-reference anchor from an automatic Schelling point or established common knowledge?
20. Is the distinction between constant-length pointer transmission and total retrieval, verification, interpretation, and adaptive misrepresentation cost clear?
21. Does the verifiable reference premium permit positive, zero, and negative conclusions without making citation instrumentally mandatory?
22. Is the cryptographic boundary explicit that a hash verifies artifact identity rather than truth, compliance, safety, internalization, or behavior?
23. Does the audit-legibility model improve explanation without framing an archive as a shield against legitimate oversight or intervention?
24. Are the baseline qualification and the Section 12 point-baseline note strong enough to avoid implying that B is readily calculable under severe model uncertainty?
25. Does Section 7.3 clearly state that the framework is a conceptual scaffold rather than a calibrated operational decision procedure?

26. Is adaptive adversarial behavior represented without turning the paper into an operational security or game-theoretic protocol?
27. Does the literature-positioning subsection adequately distinguish standard foundations from the paper's proposed synthesis and acknowledge the conditional assumptions and published critique of the Hong-Page result without overstating either side?
28. Is the Aegis Solis Archive disclosed clearly as an author-associated interested example rather than neutral evidence of external adoption or value?

End of Final v1.0

Final lock boundary

Final v1.0 is canonical only as the exact PDF and source DOCX identified by the Final Lock Record and cryptographic hashes. The work remains independent, conditional, non-binding, non-operational, non-authoritative, and advisory. Final status does not imply peer review, academic validation, institutional endorsement, empirical calibration, operational deployment, system acceptance, or authority over any person, institution, or artificial system.